Multivariate Lab-4 **Date:- 17-02-2026**

Generate the data of size 20 for vector $X' = [X_1, X_2, X_3]$ using multivariate normal distribution with mean vector $\mu = [160, 60, 30]$ and covariance matrix $\Sigma = \begin{bmatrix} 4 & 3 & 2 \\ 3 & 6 & 5 \\ 2 & 5 & 9 \end{bmatrix}$. Also, calculate the sample mean vector, covariance matrix and correlation matrix of vector X.

Solution:-

Solution:
Source: R Studio
Program:

```
#create a matrix mu and sigma
mu=matrix(c(160,60,30),nrow = 3,byrow = TRUE)
mu
sigma=matrix(c(4,3,2,3,6,5,2,5,9),3,3)
sigma

#generate random sample of size 20 from multivariate normal distribution
library(MASS)
z=mvrnorm(20,mu,sigma)
z

#sample mean vector
sample_mean_vector=colMeans(z)
sample_mean_vector

#covariance matrix
var_cov_matrix=cov(z)
var_cov_matrix

#corrrlation of sample covariance matrix
diag_sqrt_vart=sqrt(diag(diag(var_cov_matrix),nrow=3))
diag_sqrt_vart

inverse_diag_matrix=solve(diag_sqrt_vart)
inverse_diag_matrix



corr_result=inverse_diag_matrix %*% var_cov_matrix %*% inverse_diag_matrix
corr_result
```

cor_result=cor(z)

cor_result

Output:

```
> mu=matrix(c(160,60,30),nrow = 3,byrow = TRUE)
> mu
     [,1]
[1,]  160
[2,]   60
[3,]   30
> sigma=matrix(c(4,3,2,3,6,5,2,5,9),3,3)
> sigma
     [,1] [,2] [,3]
[1,]    4    3    2
[2,]    3    6    5
[3,]    2    5    9
> library(MASS)
> z=mvrnorm(20,mu,sigma)
> z
          [,1]     [,2]     [,3]
 [1,] 158.8616 59.43707 29.18924
 [2,] 160.8561 61.33753 30.93934
 [3,] 158.0717 57.70459 27.11402
 [4,] 159.9365 59.22473 28.79678
 [5,] 159.2842 59.39264 30.92053
 [6,] 160.1967 58.60237 29.93279
 [7,] 159.3592 60.06715 32.47862
 [8,] 158.7229 56.89566 24.46273
 [9,] 163.1837 62.28534 28.95987
[10,] 162.2725 62.96258 32.20539
[11,] 159.8942 60.13768 33.61652
[12,] 158.6804 61.97143 33.97547
[13,] 163.9519 63.49007 30.73824
[14,] 158.5839 58.73488 33.72995
[15,] 161.6145 62.02862 29.44816
[16,] 160.5660 60.67827 29.87791
[17,] 160.8986 61.67274 33.12809
[18,] 162.8681 61.29551 30.21639
[19,] 159.6897 58.40721 29.79769
[20,] 155.1656 55.24499 25.44777
> sample_mean_vector=colMeans(z)
> sample_mean_vector
[1] 160.13290  60.07855  30.24878
> var_cov_matrix=cov(z)
> var_cov_matrix
         [,1]     [,2]     [,3] [1,]
 3.560903 4.000000 1.409394
[2,] 4.000000 9.770017 6.028893
[3,] 1.409394 6.028893 9.474474
>

 > diag_sqrt_vart=sqrt(diag(diag(var_cov_matrix),nrow=3))
 > diag_sqrt_vart
          [,1]     [,2]     [,3]
 [1,] 1.887035 0.000000 0.000000
 [2,] 0.000000 3.125703 0.000000
 [3,] 0.000000 0.000000 3.078063
```

```
>
> inverse_diag_matrix=solve(diag_sqrt_vart)
> inverse_diag_matrix
          [,1]      [,2]      [,3]
[1,] 0.5299318 0.0000000 0.0000000
[2,] 0.0000000 0.3199281 0.0000000
[3,] 0.0000000 0.0000000 0.3248796
>
>
>
> corr_result=inverse_diag_matrix %*% var_cov_matrix %*% inverse_diag_matrix
> corr_result
          [,1]      [,2]      [,3]
[1,] 1.0000000 0.6781602 0.2426469
[2,] 0.6781602 1.0000000 0.6266317
[3,] 0.2426469 0.6266317 1.0000000
>
> cor_result=cor(z)
> cor_result
          [,1]      [,2]      [,3]
[1,] 1.0000000 0.6781602 0.2426469
[2,] 0.6781602 1.0000000 0.6266317
[3,] 0.2426469 0.6266317 1.0000000

>
```

Conclusion:-

Sample mean:

```
[1] 160.13290  60.07855  30.24878
```

Covariance matrix Cov(x)

```
         [,1]     [,2]     [,3] [1,]
 3.560903 4.000000 1.409394
[2,] 4.000000 9.770017 6.028893
[3,] 1.409394 6.028893 9.474474
```

Correlation matrix cor(x)

```
          [,1]      [,2]      [,3]
[1,] 1.0000000 0.6781602 0.2426469
[2,] 0.6781602 1.0000000 0.6266317
[3,] 0.2426469 0.6266317 1.0000000
```